MIL-B-9925C
12 February 1976
SUPERSEDING
MIL-B-9925B
24 February 1970

MILITARY SPECIFICATION

BORING MACHINES, JIG, VERTICAL, MANUAL CONTROL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manually operated vertical jig boring machines for locating, machining and measuring various work parts to precision tolerances. Machining operations include drilling, boring, reaming and light milling on parts such as jigs, dies, fixtures and production type parts.

1.2 Classification. The machines covered by this specification are of the following sizes based on table travel. The size supplied shall be as specified (see 6.2.1).

Size 914 ---- 9" by 14" Table Travel
Size 1118 ---- 11" by 18" Table Travel
Size 1216 ---- 12" by 16" Table Travel
Size 1630 ---- 16" by 30" Table Travel
Size 1828 ---- 18" by 28" Table Travel
Size 2036 ---- 20" by 36" Table Travel
Size 2236 ---- 22" by 36" Table Travel
Size 2446 ---- 24" by 46" Table Travel
Size 2848 ---- 28" by 48" Table Travel
Size 3660 ---- 36" by 60" Table Travel

2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein.

FSC 3411

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SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
FED-STD-595	Colors

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS HANDBOOK (NBS)

Handbook H-28 - Screw Thread Standard for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS (ANS)

ANS B5.1	T-Slots, Their Bolts, Nuts, and Tongues
ANS B5.10	Machine Tapers, Self-Holding and Steep Taper Series

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ANS B5.40 Spindle Noses and Tool Shanks for Horizontal Boring Machines

* ANS Z55.1 Gray Finished for Industrial Apparatus and Equipment

(Application for copies should be addressed to the American National Standards Institute, Dept. 969., 1439 Broadway, New York, NY 10018.)

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Metric Practice Guide E 380

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360 - Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

JOINT INDUSTRIAL COMMITTEE (JIC)

EGP-1 - Electrical Standard for General Purpose Machine Tools

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

* 3.2 Design. The machine shall be new and one of the manufacturer's current models. Design of the machine shall comply with the requirements herein. Each machine shall be capable of precision boring, drilling, and tapping operations to tolerances specified herein. The

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worktable and saddle arrangement shall position and feed in both the longitudinal and cross travel directions. The spindle head shall be vertically adjustable on the column. The spindle and quill shall be vertically adjustable in the spindle head. The machine design shall include all components, part and features necessary to meet the tolerance and performance requirements specified herein.

* 3.2.1 Operating controls. Operators controls for all functions of the machine shall be arranged in a location convenient to the operator's normal work position. Speeds, feeds, start/stop, clamp/unclamp, table and saddle traverse, and coolant on/off shall be controlled from the operator's control station. Controls such as pushbuttons, dials and switches, shall be functionally arranged and clearly identified to facilitate use. Emergency controls to stop all machine operations shall be provided.

* 3.2.2 Safety measures. Covers, guards, or other type safety devices shall be provided to protect the operator from hazardous moving components of the machine which can be guarded without interference with normal machine functions or operator performance. Protective devices for parts or mechanisms requiring operator attention for setting or adjustments shall be designed for easy access. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

3.2.3 Lubrication. All bearings, gears, journals, spindles, ways, and mating sliding parts shall be provided with means for lubrication. Each lubricant reservoir shall have means for determining fluid level. Each recirculating system shall include a filter which is cleanable or replaceable. Ways shall be lubricated by an automatic metered, or one-shot forced feed system. Transmission and enclosed gear trains shall be flood lubricated or equal. Motors shall have sealed-for-life lubricated bearings, or shall be equipped with grease fittings. All other parts of the machine shall be lubricated by automatic forced feed systems.

3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

* 3.3 Construction. All parts and materials used in the construction of the machine shall be new. The type of materials used shall have the necessary qualities to meet the standards and requirements specified herein for construction, safety, accuracy and performance. All parts and mechanisms subject to periodic adjustment, repair, or replacement

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shall be readily accessible. Construction of the machine shall be complete so that when installed and connected to a power supply it will be ready for operation. Unless otherwise specified (see 6.2.1), the machine may be constructed of parts manufactured to U.S. inch or International System of Units (SI) and all related printed data, plates, dials etc. shall agree.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. Welding, peening, plugging, or filling with solder or metallic paste shall not be used for reclaiming any defective part for use in the machine.

3.3.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. External surface edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

* 3.3.5 Painting. Unless otherwise specified, the machine shall be painted in accordance with the manufacturer's standard commercial practice. When specified (see 6.2.1), the final color shall be Standard Machine Tool Light Gray similar to No. 16187 Gray Gloss, FED-STD-595, or ANS Z55.1, Chip No. 70, Light Gray.

* 3.3.6 Threads. All threaded parts used in the construction of the machine shall be in accordance with applicable thread standards in Handbool H28. Leadscrew threads may be manufacturer's standard.

* 3.3.7 Dials. All dial type devices used to indicate machine functions or to select operational functions shall be legibly and permanently marked on a non-glare, contrasting finish. Each device shall be graduated to read in standard terms and units of the function for which it is used such as horsepower, units of measurement, feed rates or speeds. The devices may be mounted on the machine, control pendant, or console as appropriate and convenient to the operator's normal work station.

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3.3.8 Plates. All words on speed and feed indicating plates and instruction plates shall be in the English language, engraved, embossed or stamped in plain bold-face, easily read letters with contrasting finish.

3.3.9 Gears. All gears used in the machine and its components shall conform to the applicable requirements of AGMA 360.

* 3.4 Components. The machine shall consist primarily of a base, column, spindle head and spindle assembly, worktable, saddle, driving mechanisms, and operating controls.

3.4.1 Base. The base shall be ribbed and braced to minimize distortion, deflection and vibration. It shall form a rigid foundation for the column and the working components of the machine. When fluid reservoirs are required, they may be integral with the base or externally mounted units. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed with the base. Precision guideways shall be provided to accommodate a sliding saddle. Means for securing and leveling the machine on a foundation shall be provided around the periphery of the base.

3.4.2 Column. The column shall be of boxlike construction, internally ribbed and braced to minimize distortion, deflection and vibration under normal tool loads. The column shall have precision ways for supporting the spindle head and its related components. Gibs or other equally suitable means to adjust for wear shall be provided. Precision finished ways shall provide accurate guidance and support of the head assembly throughout its travel range.

* 3.4.3 Spindle head. The spindle head shall contain the spindle and quill assembly, depth control system, when provided, and the head elevating mechanism. The head shall move vertically on the column guideways. Positioning shall be by a handwheel, and power when provided as standard equipment, or when specified (see 6.2.1), the spindle head elevating mechanism shall be power driven and shall include automatic power clamping. Rapid traverse rates, if provided, shall be in accordance with Table I. A locking device shall be provided to clamp the head securely in position when power clamping is not required. A counterweight arrangement shall be provided to facilitate head movement.

* 3.4.4 Spindle and quill assembly. The spindle and quill assembly shall include the spindle, quill, and related components. The spindle shall rotate and be supported in the quill on precision anti-friction bearings. Unless otherwise specified, the spindle nose shall be the manufacturer's standard type. When specified (see 6.2.1), the spindle

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shall be a quick tool change type, and when specified, the spindle rotation shall be reversible. Coarse and fine hand feed shall be provided. Tool holders, adapters, or other workholding devices supplied shall conform to ANS B5.40, and be for the type spindle nose provided. Pushbutton or selector switches shall provide operator control of the spindle direction of rotation, start/stop, feed, speed, jog, and braking. Suitable devices shall be provided for selecting and indicating the spindle speeds and feeds, their range shall be in accordance with Table I.

* 3.4.5 Saddle. The saddle shall be constructed to withstand all loads imposed during operations of the machine. The saddle shall move on the guideways of the base and provide rigid support of the table. Saddle positioning shall be by sensitive handwheel, and by power when provided as standard equipment, or when power drive is required for the saddle and table (see 3.4.6). The saddle shall be provided with protective shields for the leadscrew and guideways. A saddle clamping device shall be provided to prevent saddle movement after positioning. Over-travel protection shall be provided at extremes of saddle travel. The saddle ways for supporting and guiding the worktable shall provide table travel accuracy required herein.

* 3.4.6 Worktable. The worktable shall be constructed to withstand all loads imposed during normal operations of the machine. Worktable positioning shall be by sensitive handwheels, and by power when provided as standard equipment, or when specified (see 6.2.1), the worktable and saddle shall be power driven. Rapid traverse rates, if provided, shall be in accordance with Table I. The worktable shall be equipped with protective shields for the leadscrew and guideways. Means shall be provided to prevent worktable movement after positioning. Over-travel protection shall be provided at extremes of worktable travel. A coolant trough shall be integral with the worktable with provisions for draining the coolant into the reservoir. Unless otherwise specified (see 6.2.1), the number of and location of T-slots shall be as normally provided by the manufacturer, and shall be in accordance with ANS B5.1.

* 3.4.7 Spindle drive mechanism. The spindle drive mechanism shall be electric motor drive with provisions for spindle speed changed. Speed and feed changes shall be accomplished by pushbutton, dials, switches or levers at the operator's control station.

* 3.4.8 Measuring system. Unless otherwise specified, a precision position, measuring system, of the manufacturer's standard type, shall be provided for accurately locating the table in each direction of movement and for precision measuring. The measuring system shall be graduated in the U.S. customary inch system, or in both the inch and

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International System of Units (SI). When specified (see 6.2.1), the measuring and positioning systems shall be of the type specified, shall be provided for spindle travel, and shall be graduated in the inch or SI (metric) units as required. System accuracy for table positioning shall be to ± 0.0001 inch, and spindle movement to ± 0.0005 inch or the equivalent SI unit as applicable. SI units shall be in accordance with ASTM E 380.

* 3.4.9 Work light. A work light of the built-in or adjustable arm type, shall be provided to illuminate the table work area.

3.4.10 Automatic positioning. When specified (see 6.2.1), automatic positioning shall be provided for both longitudinal and cross travel of the table. The system shall have pushbutton or selector switch control of automatic positioning, dialed in or preselected.

* 3.4.11 Automatic spindle cycle. When specified (see 6.2.1), an automatic spindle cycle shall be provided with functions including start spindle rotation while rapid advancing to workpiece (60 IPM minimum), engage feed, stop feed at preset depth and retract spindle to starting position.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machine shall meet the requirements of Table I. Options shall be supplied when specified.

3.6 Alignment and positioning accuracies. The machine alignment and positioning accuracies shall meet the tolerance of Table II.

3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system, including AC or DC drive motors, shall be designed for operation on 230/460 volt, 3 phase, 60 Hz power supply and shall be initially wired for operation on 230 volts. The electrical system shall be in accordance with the applicable requirements of JIC Electrical Standards for General Purpose Machine Tools. An identified terminal shall be provided, suitable for connecting a grounding conductor of the size specified in Table 15-1 of the JIC Standard. The main drive motor shall be the ball bearing type, shall meet the requirements of a drip-proof enclosure.

* 3.8 Coolant system. Unless otherwise specified, a flood type coolant system shall be provided. When specified (see 6.2.1), a mist system or a combination of both types shall be provided as required. The flood type system shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The coolant system shall include a

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baffle or strainer which is easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. The mist type system shall include a reservoir, filter, means for pressurizing the coolant, and means for directing and controlling the coolant.

* 3.9 Hydraulic systems. Hydraulic systems, when used, shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level. When the hydraulic system is a self-contained unit, separate from the machine, and noise level of the unit exceeds the OSHA standard, a removable noise suppression enclosure shall be provided for the unit.

3.10 Standard equipment. All special tools designed for use on the machine or accessories, and equipment normally supplied as standard equipment by the manufacturer shall be provided.

3.11 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.11.1 Elapsed time meter. The time meter shall be a digital readout, non-reset type designed to resist the effects of vibration, voltage fluctuation, temperature and humidity. The meter shall be mounted on the machine and connected to record the accumulated operating time of the machine to a total of at least 9,999.9 hours.

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TABLE I

SIZES AND CAPACITIES

(Dimensions in Inches)

CHARACTERISTIC	REQUIREMENT									
	Size 914	Size 1118	Size 1216	Size 1630	Size 1828	Size 2036	Size 2236	Size 2446	Size 2848	Size 3660
Capacity										
Table working surface, minimum, width by length	10 by 19	11 by 24	12 by 18	17 by 30	18 by 32	18 by 42	22 by 44	24 by 50	22 by 54	36 by 72
Table travel, minimum, width by length	9 by 14	11 by 18	12 by 16	10 by 30	18 by 28	20 by 36	22 by 36	24 by 46	28 by 48	36 by 60
*Optional table travel	-	-	-	-	18 by 30 or 18 by 36	-	22 by 42 or 22 by 48	24 by 48	-	-
Table to spindle nose, range	3 to 16	3-7/8 to 20	4 to 17	6 to 23	5 to 27	4 to 28	6 to 27	8 to 26	6 to 27	12 to 37
Head travel on column	9	11-15/16	9	12	15	15	14	12	14	17
Spindle quill travel	3-7/8	5	4	5	9	9	9	10	9	10
Spindle quill diameter	2-5/8	3	2-7/8	3	4-7/8	4-7/8	4-5/8	5	4-7/8	5-1/2
Spindle center to column face	10	10	9	12	16	16	18	18	18	18
Spindle center to column neck	12	13-3/4	13-3/4	21	17	21	24	26-3/4	30	44
Speeds and Feeds										
Spindle speeds, number	infinite	infinite	infinite	infinite	16	infinite	infinite	infinite	16	16
Spindle speed, range, revolutions per minute	120 to 2400	60 to 2250	100 to 2100 or 200 to 4200	30 to 1500	40 to 2000	43 to 2000	10 to 2000	30 to 1500	30 to 1800	40 to 1834
Spindle feeds, number	2	3	2	3	8	8	8	6	8	8
Spindle feed range, inches per revolution or inches per minute	0.001 to 0.003 -	0.0075, 0.0015, 0.003 -	0.0015 to 0.003 or variable 0 to 0.023	0.001, 0.003, 0.006 -	0,000 to 010 0 to 27	0.0005 to 0.015 or 0 to 27	0.0005 to 0.015 or 0 to 27	0.001, 0.003, 0.006 or 0.002, 0.006, 0.012 or variable 0 to 23	0.0005 to 0.015 or variable 0 to 27	0.0005 to 0.015 -
**Head, rapid traverse, inches per minute	-	-	40	40	20	60	-	40	60	60
**Worktable and saddle rapid traverse, inches per min.	60	60	60	50	100	60	-	50	60	60
Motors, Horsepower										
Spindle drive	2 speed 1/2 & 1	2 speed 1 & 2	Single speed 1, or 2 speed 1&2	1	5	5	5	5	5	5

* When optional sizes are required, the applicable corresponding machine characteristics such as table size apply.
** Applicable only when power operated.

Requirements are not less than those shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum. SI units when applicable shall be the nearest equivalent of the inch unit specified.

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TABLE II

ALIGNMENT TESTS
(Dimensions in inches)

REQUIREMENTS	PERMISSIBLE TOLERANCE
Concentricity of spindle nose internal bore, 3/8 inch from nose	0.0001" TIR
Concentricity of spindle, bore (runout) (proving bar in spindle nose bore)	
1/4 inch from nose	0.0001" TIR
15 inches from nose	0.0005" TIR
Worktable, trueness of travel, vertically and horizontally:	
Sizes - 914, 1118, 1216, 1218	0.0002" overall
Sizes - 1630, 1828, 1830, 1832, 1836	0.0003" overall
Sizes - 2036, 2236, 2242, 2248	0.0004" overall
Sizes - 2446, 2448, 2848, 3660	0.0005" overall
Spindle axis square with worktable surface (tramming method), largest practicable diameter of tramming circle	0.0002"/ft.
Quill motion square to worktable surface (Head locked before each reading)	
Quill movement alone	0.00015"/10"
Head movement alone	0.00015"/10"
Head and quill combined movement	0.00025" overall
Worktable front parallel with longitudinal motion	0.00015"/ft. 0.0005 overall
T-slots parallel to each other	0.001"/ft. 0.0015" overall
Longitudinal motion square to transverse motion	0.00015"/ft. 0.0003" overall
Worktable flatness (flat to concave only)	0.0001"/ft. 0.0002" overall
Worktable positioning accuracy each direction,	
Sizes 914, 1118	0.0001" overall
Other sizes	0.0002" overall

SI units, when applicable, shall be the nearest equivalent to the inch unit specified.

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3.12 Additional requirements. Performance features attachments or accessories required in addition to or in lieu of requirements herein shall be as specified by the procuring activity (see 6.2.1).

3.13 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with MIL-STD-461.

3.14 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.15 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification Number (if available)

3.16 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Table size

Power input (volts, total amps, phase, frequency)

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Contract Number

Federal Stock Number, National Stock Number, or Plant Equipment Code

Date of manufacture

U S

3.17 Technical data. Data shall be furnished as specified (see 6.2.2).

3.18 Workmanship. Workmanship of the machine and accessories shall be equal to that of the manufacturer's current commercial machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4, and tests in 4.5.1 through 4.5.3. Failure to pass the examination or any test shall be cause for disapproval of the First Article.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.3, and the inspection in 4.6. Failure of an item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined for compliance with the requirements of this specification in 3.2 through 3.5 and 3.7 through 3.18.

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4.5 Tests.

4.5.1 Operational tests. Each machine shall be operated in accordance with the supplier's standard operating test procedures for warm-up and run-off checks. Run-off checks of all mechanical functions, and electrical and measuring systems shall be included. The checks shall include random spindle speed changes and feed-rate changes, and rapid travel of each unit so equipped.

* 4.5.2 Performance tests. The following test shall be performed to demonstrate the machine's capability of accurately positioning, boring and relocating. Drill and bore 3 holes, 3/4 inch diameter by 3/4 inch deep in a right angle pattern, in a metal plate not less than 1 inch thick. The plate shall be of a material chosen by the supplier and both faces may be surface ground prior to the test. Counter bore the holes 1 inch by 1/8 inch depth. The dimensions between the bored holes shall be spaced to utilize as much of the normal working range of the machine as practical, allowing for set-up devices. Using a dial indicator and the measuring system, relocate the bored holes, (two in-line on each axis). The machine shall accurately relocate each hole to within plus or minus 0.00025 inch center to center, counter bores concentric with bored holes within 0.0002 TIR.

* 4.5.3 Alignment and positioning test. The machine shall be checked to determine that component alignment and positioning accuracies are within the tolerances of 3.6, Table II, when checked in accordance with the manufacturer's standard test procedures. Test instruments used may be the manufacturer's choice provided they have an accuracy traceable to the Bureau of Standards.

4.5.4 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.13.

4.6 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. The required level of preservation-packaging and packing, and any special marking requirements shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. The jig boring machines covered by this specification are intended for use in any toolroom or production shop to locate, drill, bore, tap or measure parts to precision tolerances.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.

b. Size required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.4).

e. Specify one system of measurement if required (see 3.3).

f. Painting, if different, machine tool light gray if required (see 3.3.5).

g. Power driven spindle head elevation if required (see 3.4.3).

h. Spindle requirements if different (see 3.4.4).

i. Power driven table if required, number and position of table T-slots, if different (see 3.4.6).

j. Measuring system type if other than manufacturer's standard, and measuring system for spindle travel, if required, inch or SI system as required (see 3.4.8).

k. Automatic positioning, if required (see 3.4.10).

1. Automatic spindle cycles, if required (see 3.4.11).

m. Sizes and capacities, if different, and options, if required (see 3.5, Table I).

n. Electrical system, if different and voltage for initial wiring (see 3.7).

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o. Specify coolant system type(s) if different (see 3.8).

p. Optional equipment, as required (see 3.11).

q. Additional features or equipment, if required (see 3.12).

r. Electromagnetic interference control, if required (see 3.13).

s. Type fungus proofing, if required (see 3.14).

t. Lubrication chart, if different (see 3.15).

u. Nameplate, if different (see 3.16).

v. First Article inspection, if different (see 4.2).

w. Quality conformance inspection, if different (see 4.3).

x. Electromagnetic interference control test requirements, if different (see 4.5.4).

y. Required level of preservation-packaging, packing and special marking requirements (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.18). Consideration should be given to items needed prior to delivery of the machine tool.

6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

6.4 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

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6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH
DSA-GS

User Activities:
Army - WV
Navy - MC

Preparing Activity:
DSA - IP

Project Number:
3411-0051

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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